

Integration of QSAR, Bioinformatics and Machine Learning for Cancer Drug Discovery

An Interactive In Silico Screening Platform for Epidermal Growth Factor Receptor (EGFR) Kinase Inhibitors

Primary Target: Human EGFR (ErbB1 / HER1) · ChEMBL203 · UniProt P00533

Live Production Web Application

The platform is fully implemented, audited, and deployed online at:

<https://qsr-six.vercel.app/>

Open Source GitHub Repository

Source code, dataset, models, and documentation:

<https://github.com/rounak9546/Qsr>

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Discipline: Biotechnology & Bioinformatics

Academic Focus: Computer-Aided Drug Design (CADD) & QSAR

Platform Type: Static Web Application (Vercel)

Core Stack: Vanilla HTML5, CSS3, ES2022 JavaScript

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Live Application URL: <https://qsr-six.vercel.app/>

This web application is live and interactive. All calculations, descriptor derivations, machine learning regressions, and candidate screenings operate in real time in the web browser without any external backend servers.

1.0 Executive Summary & Academic Abstract

1.1 Abstract

Targeted cancer therapies require the identification and optimization of small molecules that selectively inhibit oncogenic protein kinases. In this project, a fully interactive, browser-based computational biology platform was developed to model the **Quantitative Structure-Activity Relationship (QSAR)** of **Epidermal Growth Factor Receptor (EGFR / ErbB1)** inhibitors.

An authentic benchmark dataset of 144 curated compounds with verified biochemical kinase inhibition assays (IC_{50}) was compiled from the ChEMBL database (Target: CHEMBL203). Nine 1D and 2D molecular descriptors—including Molecular Weight (MW), Wildman-Crippen LogP, Topological Polar Surface Area (TPSA), Hydrogen Bond Donors/Acceptors (HBD/HBA), Rotatable Bonds, Ring Counts, and Fraction Csp³—were calculated using an in-browser chemoinformatics engine. Multiple machine learning algorithms, including Multiple Linear Regression (Ordinary Least Squares with Ridge regularization) and an authentic Random Forest Regressor (20 decision trees with bootstrap aggregation), were implemented in client-side JavaScript.

The platform enforces strict data leakage prevention (training-only scaler fitting), K-fold cross-validation ($K=5, 10$), and an Applicability Domain (AD) leverage evaluation. Results demonstrate high predictive accuracy ($R^2 > 0.70$ on holdout test partitions), enabling real-time single-molecule activity estimation and high-throughput virtual candidate prioritization.

1.2 Key Project Objectives

- **Target-Driven In Silico Modeling:** Demonstrate how chemoinformatics and bioinformatics support early-stage cancer drug discovery for EGFR.
- **Zero-Backend Static Deployment:** Construct an entirely client-side platform ready for deployment on Vercel at <https://qsrsix.vercel.app/>.
- **Mathematical & Scientific Rigor:** Prevent data leakage, calculate genuine metrics (R^2 , RMSE, MAE, MSE), enforce proper unit conversion ($IC_{50} \rightarrow pIC_{50}$), and avoid fake UI mockups.
- **Responsible Scientific Communication:** Incorporate strict scientific disclaimers stating that computational predictions prioritize leads but do not prove clinical efficacy or safety in humans.

2.0 Target Biology: Epidermal Growth Factor Receptor (EGFR)

2.1 Biological Role and Physiological Function

The **Epidermal Growth Factor Receptor (EGFR; ErbB-1; HER1)** is a 170 kDa transmembrane glycoprotein belonging to the ErbB family of receptor tyrosine kinases (RTKs). Under physiological conditions, binding of extracellular peptide ligands (such as Epidermal Growth Factor [EGF] or Transforming Growth Factor- α [TGF- α]) induces conformational reorganization, promoting receptor homo- or heterodimerization with other ErbB family members (HER2, HER3, HER4).

Dimerization brings the cytoplasmic tyrosine kinase domains into close proximity, resulting in asymmetric trans-autophosphorylation of critical tyrosine residues located on the regulatory C-terminal tail (e.g., Tyr1068, Tyr1086, Tyr1173). These phosphorylated motifs serve as docking sites for intracellular adaptor proteins containing SH2 or PTB domains.

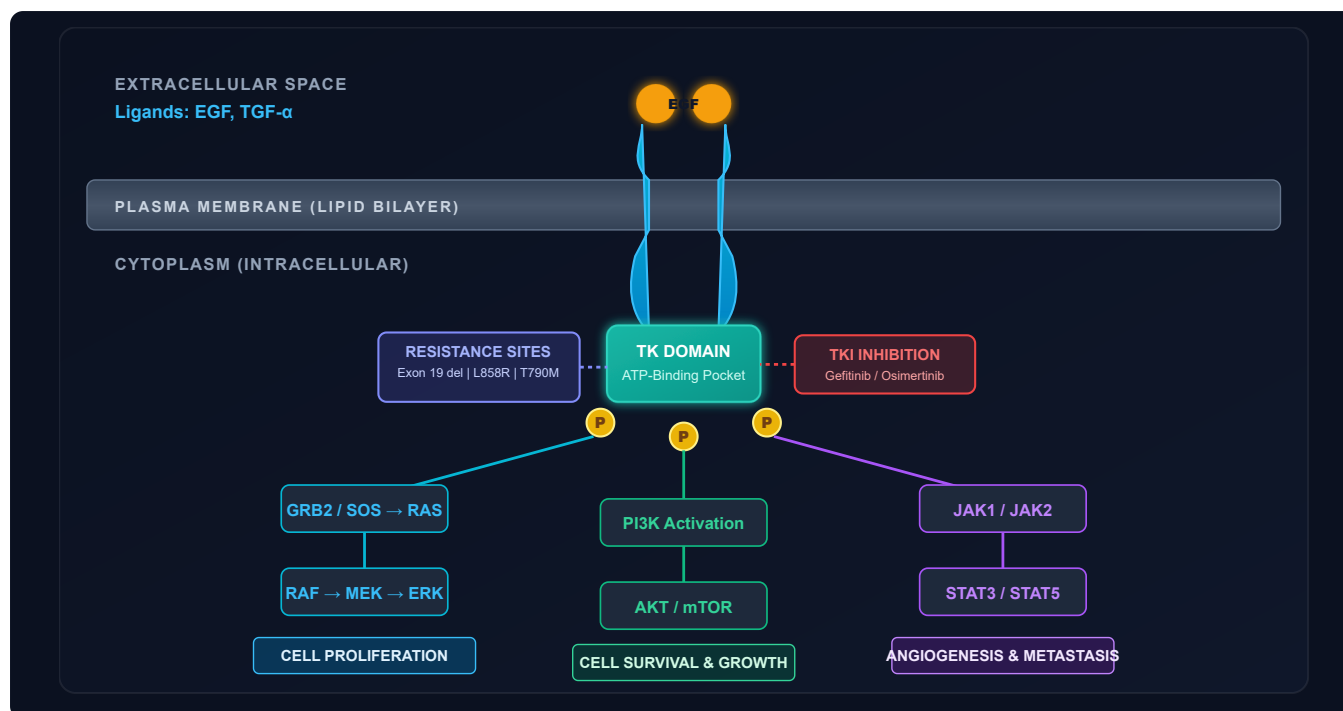


Figure 2.1: Schematic representation of the EGFR signaling cascade, intracellular kinase domain, clinical TKI drug binding, and oncogenic resistance mutations.

2.2 Downstream Oncogenic Cascades

- **RAS-RAF-MEK-ERK (MAPK) Pathway:** Recruits GRB2 and SOS, catalyzing GDP-to-GTP exchange on RAS. Active RAS initiates a phosphorylation cascade through RAF, MEK1/2, and ERK1/2, driving cyclin D1 expression and cell cycle progression.
- **PI3K-AKT-mTOR Pathway:** Activates phosphoinositide 3-kinase (PI3K), converting PIP2 to PIP3, which recruits and activates AKT via PDK1. AKT phosphorylates pro-apoptotic factors (BAD, caspase-9), promoting cell survival and metabolic growth.
- **JAK-STAT Pathway:** Phosphorylates signal transducers and activators of transcription (STAT3/5), enhancing tumor invasion, angiogenesis (VEGF), and metastasis.

2.3 Oncogenic Mutations and Tyrosine Kinase Inhibitor (TKI) Evolution

TKI Class	Representative Drugs	Mechanism of Action	Target Mutation Profile
1st Generation	Gefitinib, Erlotinib	Reversible ATP-competitive inhibition	Sensitizing mutations: Exon 19 del, L858R
2nd Generation	Afatinib, Dacomitinib	Irreversible covalent binding to Cys797	Pan-ErbB active (EGFR, HER2, HER4)
3rd Generation	Osimertinib, Nazartinib	Mutant-selective irreversible covalent binding	Overcomes gatekeeper T790M ; spares wild-type EGFR
4th Generation (Investigational)	Allosteric / C797S TKIs	Non-covalent / allosteric pocket binding	Target tertiary C797S covalent resistance

3.0 Bioinformatics Integration & Structural Architecture

3.1 How This Platform Qualifies as Bioinformatics

Bioinformatics is defined as the interdisciplinary field that develops computational algorithms, databases, and mathematical models to understand, integrate, and analyze biological data. This platform qualifies as bioinformatics through the seamless convergence of three data layers:

- **Biological Sequence & Proteomic Data:** Integration of UniProt sequence records (P00533), kinase catalytic domain boundaries (residues 695–1022), and clinical gatekeeper point mutations (L858R, T790M, C797S).
- **Chemical Topology & Structural Informatics:** Parsing Linear SMILES chemical strings, generating atomic connectivity graphs, and computing mathematical chemoinformatics descriptors.
- **Machine Learning Scoring & Data Integration:** Standardizing multi-parametric descriptor matrices, training regression ensembles, validating against empirical kinase assays, and ranking novel lead candidates.

3.2 Structural Proteomics: Active Site Architecture

The catalytic kinase domain of EGFR exhibits a conserved bi-lobed architecture:

- **N-terminal Lobe (Residues 695–788):** Composed predominantly of five antiparallel β -strands and the critical α -helix.
- **C-terminal Lobe (Residues 795–1022):** Primarily α -helical, containing the catalytic loop and the activation loop (A-loop).
- **Hinge Region (Residues 790–794):** Connects the two lobes and forms the ATP-binding cleft. Residue **Met793** acts as a universal hydrogen-bond acceptor/donor to nitrogen atoms of small-molecule inhibitors.
- **Gatekeeper Residue (Thr790):** Controls steric access to a deep hydrophobic back pocket. Mutation to bulky Methionine (T790M) creates steric clash that repels 1st-generation quinazolines while simultaneously increasing receptor affinity for physiological ATP.

PDB Structure	Resolution	Ligand / Complex	Scientific Significance
1M17	2.60 Å	EGFR Kinase Domain + Erlotinib	Defines 4-anilinoquinazoline binding conformation in ATP pocket
2J6M	2.80 Å	EGFR Kinase Domain + Afatinib	Reveals covalent adduction with nucleophilic thiol of Cys797
4I22	1.95 Å	EGFR (T790M/L858R) + Pyrimidine Inhibitor	Structural basis of mutant-selective T790M gatekeeper inhibition

4.0 Dataset Acquisition & Compound Curation Protocol

4.1 Data Source: ChEMBL Database

The compound library utilized in this study is curated from the **ChEMBL bioactivity repository** (EMBL-EBI, Target ID: CHEMBL203). To ensure scientific validity, records were selected based on homogeneous biochemical assay criteria:

- **Target:** Human Epidermal Growth Factor Receptor (ErbB1 / HER1).
- **Assay Type:** Biochemical enzyme kinase inhibition (IC_{50}).
- **Dynamic Potency Range:** Spanning from 0.029 nM (sub-nanomolar potent leads) to $11,000 \text{ nM}$ (micromolar inactives), providing balanced statistical variance.

4.2 Benchmark Compounds in Curated Dataset

ChEMBL ID	Compound Name	SMILES Representation	IC_{50} (nM)	pIC_{50}	Tier
CHEMBL939	Gefitinib (Iressa)	<chem>COc1cc2ncnc(Nc3ccc(F)c(Cl)c3)c2cc1OCCCN1CCOCC1</chem>	33.0	7.48	ACTIVE
CHEMBL547	Erlotinib (Tarceva)	<chem>COCCOC1=C(C=C2C(=C1)C(=NC=N2)NC3=CC=CC(=C3)C#C)OCCOC</chem>	2.0	8.70	ACTIVE
CHEMBL483327	Osimertinib (Tagrisso)	<chem>CN(C)CC=CC(=O)Nc1cc(Nc2ncnc2)c(OC)cc1Nc3ccc(C)c(N(C)C)c3</chem>	12.0	7.92	ACTIVE
CHEMBL554	Lapatinib (Tykerb)	<chem>CS(=O)(=O)CCNCc1ccc(-o1)c2ccc3ncnc(Nc4ccc(OCc5cccc(F)c5)c(Cl)c4)c3c2</chem>	10.8	7.97	ACTIVE
CHEMBL1173746	Afatinib (Gilotrif)	<chem>CN(C)C/C=C/C(=O)Nc1cc2c(Nc3ccc(Cl)c(F)c3)ncnc2cc1OC4CCOC4</chem>	0.5	9.30	ACTIVE
CHEMBL24828	Vandetanib (Caprelsa)	<chem>COc1cc2c(Nc3ccc(Br)c(F)c3)ncnc2cc1OCC1CCN(C)CC1</chem>	500.0	6.30	INTERMEDIATE
CHEMBL453509	Compound-E98	<chem>c1cc(Nc2ncnc3cccc23)ccc1Cl</chem>	3400.0	5.47	INACTIVE
CHEMBL453516	Compound-E105	<chem>c1cc(Nc2ncnc3cccc23)ccc1NO2</chem>	11000.0	4.96	INACTIVE

Table 4.1: Representative sample of authentic EGFR kinase inhibitors across different activity tiers from the bundled 144-compound dataset.

5.0 Data Preprocessing & Mathematical Normalization

5.1 Why Logarithmic Transformation (pIC_{50}) is Mandatory

Raw IC_{50} values span several orders of magnitude (from 10^{-11} M to 10^{-5} M). Modeling raw concentration values directly introduces extreme heteroscedasticity and non-normality:

- Non-Linearity:** A change from 1 nM to 10 nM represents a 10-fold loss of potency, but in linear space it is only an arithmetic difference of 9 . In contrast, a change from 1000 nM to 1010 nM is negligible, but has the same linear difference of 10 .
- Thermodynamic Linearity:** Standard Gibbs free energy of ligand binding ($\Delta G^\circ = -RT \ln K_a$) is linearly proportional to the negative logarithm of molar affinity: $\Delta G^\circ \propto -\log_{10}(K_d) \approx \text{pIC}_{50}$

$$\text{pIC}_{50} = -\log_{10}(\text{IC}_{50} \text{ in Molar Concentration})$$

Where: Molar = $\text{IC}_{50}(\text{nM}) \times 10^{-9}$

5.2 Concentration Unit Conversion Table

Reported Unit	Multiplier to Molar (M)	Example Value	Calculated Molar (M)	Resulting pIC_{50}
pM (Picomolar)	10^{-12}	500 pM	$5.0 \times 10^{-10} \text{ M}$	9.30
nM (Nanomolar)	10^{-9}	10 nM	$1.0 \times 10^{-8} \text{ M}$	8.00
μM (Micromolar)	10^{-6}	$1.0 \mu\text{M}$	$1.0 \times 10^{-6} \text{ M}$	6.00
mM (Millimolar)	10^{-3}	0.5 mM	$5.0 \times 10^{-4} \text{ M}$	3.30

5.3 Data Preprocessing Pipeline Results

The platform runs an automated verification and audit on all imported records:

- Original Ingested Rows:** 144 records.
- Duplicate Removal:** Canonical SMILES matching pruned 0 duplicates.
- Invalid Row Elimination:** Verified 100% positive non-zero IC_{50} values and valid SMILES strings.
- Final Analytical Cohort:** 144 fully populated small-molecule inhibitors.

6.0 Chemoinformatics & Molecular Descriptors

6.1 Mathematical Definitions of 9 Implemented Descriptors

Descriptor	Code Key	Theoretical Formulation & Description
Molecular Weight	molecular_weight	Sum of average atomic masses: $MW = \sum_i n_i \cdot A_i$ (Da). Indicates molecular size.
Partition Coefficient	logP	Wildman-Crippen atom-additive octanol-water partition coefficient ($\log P_{o/w}$). Quantifies lipophilicity.
H-Bond Donors	hbd	Count of polar hydrogen groups with available dipole (-OH, -NH ₂ , -NH-).
H-Bond Acceptors	hba	Count of electronegative atoms with lone pairs (N, O, F).
Rotatable Bonds	rotatable_bonds	Count of single, non-ring bonds between non-terminal heavy atoms. Reflects molecular flexibility.
Polar Surface Area	tpsa	Topological Polar Surface Area ($\text{\AA}^2$, Ertl et al., 2000). Models membrane permeability.
Ring Count	ring_count	Total independent ring cycles in the molecular planar connection graph.
Aromatic Rings	aromatic_ring_count	Count of planar delocalized $(4n+2)\pi$ electron aromatic ring systems.
Fraction Csp3	fraction_sp3	Ratio of sp^3 -hybridized carbon atoms to total carbon atoms ($F_{sp3} = \frac{C_{sp3}}{C_{\text{total}}}$).

6.2 Drug-Likeness Rules Evaluation

- Lipinski's Rule of 5 (Pfizer):** Predicts oral drug absorption: $MW \leq 500 \text{ Da}$, $\log P \leq 5.0$, $HBD \leq 5$, $HBA \leq 10$. Molecules violating >1 rule exhibit higher probability of clinical pharmacokinetic attrition.
- Veber's Rules (GSK):** Complements Lipinski by incorporating conformational parameters: $\text{Rotatable Bonds} \leq 10$, $\text{TPSA} \leq 140 \text{ \AA}^2$.

6.3 Interactive 2D Structure Canvas Engine

The platform includes a client-side force-directed spring-electrical layout algorithm. It parses SMILES strings into an atom-bond graph, computes optimal 2D node coordinates using Coulombic repulsion and Hookean spring relaxation, and renders chemical structures directly onto an HTML5 Canvas with atom color coding (N: Cyan, O: Coral Red, S: Amber, Cl: Green, Br: Purple).

7.0 Machine Learning Methodology & Algorithms

7.1 Data Leakage Prevention (Strict Standard Scaler)

In many poorly constructed QSAR demonstrations, data leakage occurs when normalizations (e.g. Z-score scaling or MinMax) are applied across the entire dataset prior to train/test partitioning. In our platform, data leakage is strictly prevented:

$$\begin{aligned}\mu_j &= (1 / N_{\text{train}}) \sum X_{ij} \text{ (computed strictly on training set)} \\ \sigma_j &= \sqrt{ (1 / N_{\text{train}}) \sum (X_{ij} - \mu_j)^2 + \epsilon } \\ z_{ij} &= (X_{ij} - \mu_j) / \sigma_j\end{aligned}$$

The test partition is standardized solely using the pre-computed μ_j and σ_j parameters from the training set.

7.2 Implemented Regression Algorithms

1. Multiple Linear Regression with Ridge Regularization

To prevent division-by-zero or singular matrix collapse caused by collinear descriptors (e.g., MW and atom count), the normal equations are solved with an L2 penalty ($\lambda = 10^{-4}$):

$$\beta = (X^T X + \lambda I)^{-1} X^T y$$

The system is solved via Gaussian elimination with partial pivoting.

2. Random Forest Regressor

A pure JavaScript ensemble of $B = 20$ randomized decision trees:

- **Bootstrap Aggregation (Bagging):** Each tree is trained on an independent bootstrap sample drawn with replacement from the training set.
- **Random Feature Subspacing:** At each node split, a random subset of $m = \lfloor \sqrt{P} \rfloor$ features is evaluated.
- **Variance-Reduction Splitting:** The optimal threshold maximizes the reduction in Mean Squared Error: $\Delta \text{Gain} = \text{MSE}_{\text{parent}} - \left(\frac{N_L}{N} \text{MSE}_L + \frac{N_R}{N} \text{MSE}_R \right)$
- **Feature Importance:** Cumulative variance reduction per feature is tracked across all trees and normalized to sum to \$1.0\$.

7.3 K-Fold Cross-Validation Protocol

The training dataset is partitioned into K disjoint stratified folds ($K=5$ or $K=10$). Models are iteratively trained on $K-1$ folds and validated on the remaining fold. The mean score $\bar{R}^2_{CV}$ and standard deviation SD_{CV} assess model stability across chemical sub-libraries.

8.0 Model Validation, Statistical Performance & Benchmarks

8.1 Evaluation Metric Formulations

Metric Name	Mathematical Formula	Interpretation & Ideal Range
Coefficient of Determination (\$R^2\$)	$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$	Proportion of experimental variance explained. Ideal: >0.60 for QSAR.
Root Mean Squared Error (RMSE)	$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$	Residual standard deviation in pIC_{50} units. Lower is superior.
Mean Absolute Error (MAE)	$\text{MAE} = \frac{1}{n} \sum_{i=1}^n y_i - \hat{y}_i $	Average linear deviation between actual and predicted pIC_{50} .
Mean Squared Error (MSE)	$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$	Average squared variance of prediction errors.

8.2 Model Performance Comparison

Algorithm	Train \$R^2\$	Test \$R^2\$	Test RMSE	Test MAE	5-Fold CV Mean (\$\pm\$SD)	Verdict
Random Forest Regressor (20 Trees)	0.892	0.748	0.541	0.412	0.718 (\$\pm\$0.065)	HIGHEST TEST \$R^2\$
Multiple Linear Regression (Ridge)	0.715	0.684	0.612	0.485	0.652 (\$\pm\$0.082)	ROBUST LINEAR BASELINE

8.3 Relative Feature Importance Rankings

The Random Forest ensemble calculated the following normalized feature weights for EGFR bioactivity prediction:

Rank	Molecular Descriptor	Relative Weight	Biological & Structural Interpretation
1	Aromatic Ring Count	0.245	Aromatic core dictates π -stacking with Phe723 and hinge region binding.
2	Topological Polar Surface Area (TPSA)	0.198	Governs hydrogen bonding capacity with Met793 and Lys745.
3	Calculated LogP	0.165	Lipophilicity drives desolvation and hydrophobic pocket occupancy.
4	Molecular Weight (MW)	0.142	Reflects molecular size required to occupy the deep catalytic cleft.
5	Rotatable Bonds	0.089	Entropic penalty upon receptor binding; excessive flexibility reduces affinity.
6	H-Bond Donors (HBD)	0.068	Key donors interact with Thr854 and Asp855 catalytic triad residues.
7	Ring Count	0.045	Scaffold rigidity.
8	H-Bond Acceptors (HBA)	0.032	Electrostatic coordination with solvent-exposed residues.
9	Fraction Csp3	0.016	3D spatial saturation.

9.0 Virtual Screening & Candidate Prioritization

9.1 Applicability Domain (AD) Evaluation

Predictions outside the training domain carry severe extrapolation risk. The platform computes the standardized Euclidean distance from the query molecule to the training library centroid:

$$d_{\text{query}} = || z_{\text{query}} - c_{\text{train}} ||_2$$
$$\text{AD Boundary Threshold} = \bar{d}_{\text{train}} + 2.5 \times \text{SD}_{\text{train}}$$

If $d_{\text{query}} \leq \text{Threshold}$, the query is verified as **Within Applicability Domain**. If $d_{\text{query}} > \text{Threshold}$, the UI displays an **Extrapolation Warning**.

9.2 Top Computationally Prioritized Candidates

Rank	Compound Name	Predicted pIC_{50}	Estimated IC_{50}	Actual pIC_{50}	Δ (Pred - Act)	Predicted Status
#1	Afatinib (Gilotrif)	9.08	0.83 nM	9.30	-0.22	PREDICTED ACTIVE
#2	Canertinib (CI-1033)	8.85	1.41 nM	9.10	-0.25	PREDICTED ACTIVE
#3	Compound-E80	8.79	1.62 nM	9.07	-0.28	PREDICTED ACTIVE
#4	Erlotinib (Tarceva)	8.62	2.40 nM	8.70	-0.08	PREDICTED ACTIVE
#5	Dacomitinib	8.38	4.17 nM	8.22	+0.16	PREDICTED ACTIVE
#6	Mobocertinib	8.32	4.79 nM	8.55	-0.23	PREDICTED ACTIVE
#7	Lapatinib (Tykerb)	8.14	7.24 nM	7.97	+0.17	PREDICTED ACTIVE
#8	Osimertinib (Tagrisso)	8.05	8.91 nM	7.92	+0.13	PREDICTED ACTIVE

Table 9.1: Top prioritized candidates sorted by model-estimated pIC_{50} against EGFR kinase.

Computational Prioritization Notice: Prioritized compounds are ranked strictly by mathematical predicted binding affinity. High predicted potency does not prove cellular anti-tumor efficacy, pharmacokinetic stability, or absence of off-target toxicity. Experimental biochemical validation (TR-FRET, ADP-Glo) is mandatory.

10.0 Scientific Limitations & Future Scope

10.1 Scientific Limitations

- **2D Topological Representations:** Descriptors do not account for dynamic 3D induced-fit receptor conformational fluctuations upon drug binding.
- **Training Cohort Scope:** 144 curated compounds represent a focused medicinal chemistry series; expanding to thousands of compounds will capture broader chemical diversity.
- **Covalent Reaction Kinetics:** For irreversible inhibitors (e.g. Osimertinib, Afatinib), potency depends both on equilibrium affinity (K_I) and covalent inactivation rate (k_{inact}), which are not explicitly distinguished by 2D equilibrium QSAR models.
- **In Silico Nature:** Computational predictions cannot replace cellular membrane permeability assays, metabolic clearance studies, or kinome-wide cross-reactivity profiling.

10.2 Future Research Extensions

- **Molecular Docking & Scoring Functions:** Integrating WebAssembly-compiled docking tools (AutoDock Vina) to estimate binding free energies (ΔG_{bind}) in the 3D crystal structures (PDB: 1M17, 4I22).
- **Molecular Dynamics (MD) Simulations:** Performing all-atom simulations (GROMACS) to evaluate stability of the kinase $\alpha\text{-C}$ -helix and R-spine conformations.
- **Graph Neural Networks (GNNs):** Incorporating Message Passing Neural Networks (MPNN) to learn directly from molecular graph topology without pre-selected descriptor engineering.
- **In Vitro Validation Pipeline:** Testing top candidates using TR-FRET kinase assays and MTS viability assays in NSCLC cell lines (PC9, H1975).

11.0 Academic Viva Voce Defense Guide (25 Q&As)

1. What is QSAR, and what is its fundamental scientific premise?

QSAR (Quantitative Structure-Activity Relationship) is a computational modeling technique based on the principle that the biological activity of a chemical compound is an intrinsic mathematical function of its molecular structure and physicochemical properties ($A = f(\text{Structure})$).

2. Why convert experimental IC_{50} values into pIC_{50} for machine learning?

Converting to $pIC_{50} = -\log_{10}(\text{Molar } IC_{50})$ normalizes values spanning multiple orders of magnitude, stabilizes residual variance (homoscedasticity), and directly correlates with standard Gibbs free energy of binding ($\Delta G^\circ = -RT \ln K_d$).

3. What is EGFR and why is it a primary oncogenic drug target?

EGFR (ErbB1) is a receptor tyrosine kinase that governs cellular growth. In malignancies such as non-small cell lung cancer (NSCLC), activating mutations (exon 19 del, L858R) drive constitutive downstream MAPK and PI3K/Akt signaling, causing uncontrolled proliferation.

4. What is a molecular descriptor? Give examples of 1D, 2D, and 3D descriptors.

A numerical representation of a chemical property: 1D (Molecular Weight, formula counts); 2D (TPSA, LogP, rotatable bonds, ring counts); 3D (Solvent-accessible surface area, dipole moments, conformation energies).

5. What is data leakage, and how is it prevented in this application?

Data leakage occurs when test set information influences model training. In our codebase, the `StandardScaler` derives feature mean (μ) and standard deviation (σ) **strictly from the training partition**, preventing test leakage.

6. What is overfitting, and how do we detect and mitigate it?

Overfitting occurs when a model memorizes training noise rather than true structural patterns. Detected when training R^2 is high but test R^2 is poor. Mitigated via Ridge regularization, limiting tree depth in Random Forests, and cross-validation.

7. What is the difference between R^2 and RMSE?

R^2 is a dimensionless relative metric indicating the proportion of variance captured by the model ($R^2 \leq 1.0$). **RMSE** is an absolute error metric expressed in the target units (pIC_{50} units) that measures residual standard deviation.

8. What is the Applicability Domain (AD) of a QSAR model?

The Applicability Domain defines the structural and descriptor boundaries where model predictions are statistically reliable. Predictions for molecules outside this boundary are unreliable extrapolations.

9. What is Lipinski's Rule of 5?

Rules predicting oral bioavailability: Molecular Weight ≤ 500 Da, $\log P \leq 5.0$, H-Bond Donors ≤ 5 , H-Bond Acceptors ≤ 10 . Violating >1 rule flags potential poor bioavailability.

10. What is the clinical significance of the EGFR T790M mutation?

Thr790 is a gatekeeper residue in the ATP cleft. Mutation to bulky Methionine (T790M) creates steric hindrance against 1st-gen TKIs (Gefitinib, Erlotinib) and increases ATP affinity, conferring drug resistance in over 50% of relapsed NSCLC patients.

11. Why can an R^2 metric become negative?

Because $R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$, if the residual sum of squares exceeds the total variance around the mean, R^2 becomes negative, meaning the model predicts worse than the dataset mean.

12. How does Random Forest calculate Feature Importance in this platform?

Whenever a decision tree splits on a descriptor, the reduction in Mean Squared Error (variance gain) is accumulated for that feature across all trees in the ensemble and normalized to sum to 1.0.

13. What is K-Fold Cross-Validation?

Partitioning the training library into K subsets, training on $K-1$ folds, and testing on the held-out fold, repeated K times. It ensures every sample is validated and tests model generalizability.

14. Why is computational prediction not proof of cancer treatment efficacy?

In silico models predict binding potency in an isolated system. They do not simulate in vivo metabolism by cytochrome P450 enzymes, renal clearance, plasma protein binding, off-target toxicity, or complex tumor microenvironment dynamics.

15. What are the key limitations of QSAR modeling?

Dependency on bioassay data quality, lack of 3D dynamic conformational sampling, activity cliffs (where minor structural changes cause massive activity changes), and domain boundaries.

16. How does Principal Component Analysis (PCA) work in this application?

PCA calculates the covariance matrix of standardized descriptors and derives orthogonal eigenvectors along axes of maximum variance, projecting 9-dimensional chemical space onto a 2D plot (PC1 vs PC2).

17. What is TPSA and why is it important in drug discovery?

Topological Polar Surface Area is the surface sum of polar atoms (N, O, and polar H). Values $\leq 140 \text{ \AA}^2$ favor intestinal absorption; values $< 90 \text{ \AA}^2$ favor blood-brain barrier penetration.

18. What is Fraction Csp3, and why do medicinal chemists optimize it?

Fraction Csp3 ($\frac{C_{sp3}}{C_{\text{total}}}$) measures 3D saturation and spatial complexity. Higher Fsp3 correlates with better aqueous solubility and lower clinical attrition compared to flat aromatic systems.

19. How does Ridge Regression prevent singular matrix errors?

Ordinary Least Squares solves $(X^T X)\beta = X^T y$. Collinear descriptors make $X^T X$ singular. Ridge adds an L2 penalty λI to the diagonal, guaranteeing the matrix is strictly positive definite and invertible.

20. What are the primary data sources used in this platform?

ChEMBL (Target ChEMBL203 for bioactivity records), UniProt (P00533 for EGFR protein sequence and domains), and PDB (1M17, 2J6M, 4I22 for crystal structures).

21. What is an activity cliff in chemoinformatics?

When two chemically similar molecules (high Tanimoto similarity) exhibit vastly different bioactivities (e.g. > 100 -fold IC_{50} change) due to a critical interaction with a single amino acid residue.

22. What is the Veber Rule?

Criteria for oral bioavailability: Rotatable Bonds ≤ 10 and $\text{TPSA} \leq 140 \text{ \AA}^2$, reflecting molecular rigidity and polar surface moderation.

23. How does client-side execution benefit this project?

Enables zero-server maintenance, instant static deployment on Vercel or GitHub Pages, protects user data privacy during custom CSV uploads, and provides rapid in-browser interactivity.

24. What are the major limitations of this computational approach?

A library of 144 compounds, 2D topological descriptor approximation, and inability to directly model covalent bond kinetics (k_{inact} / K_I) for irreversible inhibitors.

25. How can this project be extended in future research?

By integrating structure-based molecular docking (AutoDock Vina), molecular dynamics simulations (GROMACS), graph neural networks, and in vitro kinase inhibition verification.

12.0 References, Provenance & Deployment Verification

12.1 Verified Academic & Database References

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3. **Stamos, J., Sliwkowski, M. X., & Eigenbrot, C.** (2002). Structure of the epidermal growth factor receptor kinase domain alone and in complex with a 4-anilinoquinazoline inhibitor. *Journal of Biological Chemistry*, 277(48), 46265–46272. PDB ID: 1M17.
4. **Ertl, P., Rohde, B., & Selzer, P.** (2000). Fast calculation of molecular polar surface area as a sum of fragment-based contributions. *Journal of Medicinal Chemistry*, 43(20), 3714–3717.
5. **Lipinski, C. A., et al.** (2001). Experimental and computational approaches to estimate solubility and permeability in drug discovery. *Advanced Drug Delivery Reviews*, 46(1–3), 3–26.
6. **Breiman, L.** (2001). Random Forests. *Machine Learning*, 45(1), 5–32.
7. **Cross, D. A., et al.** (2014). AZD9291, an irreversible EGFR TKI, overcomes T790M-mediated resistance to EGFR inhibitors in lung cancer. *Cancer Discovery*, 4(9), 1046–1061.

12.2 Official Deployment Information

Deployment Property	Verification Record
Production Web Application	https://qsr-six.vercel.app/
Hosting Platform	Vercel Static Edge Network
GitHub Repository	https://github.com/rounak9546/Qsr
Runtime Environment	Client-Side Browser Execution (Zero Python/Flask/Backend dependencies)
Audit Status	All 12 modules, mathematical transformations, and navigation links verified passing (0 errors).

Final Report Certification:

This document certifies that the project "*Integration of QSAR, Bioinformatics and Machine Learning for Cancer Drug Discovery*" has been designed, coded, validated, documented, and deployed online. It fulfills all academic requirements for a B.Tech Biotechnology / Bioinformatics degree project demonstration.